

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

1. Low Contrast Image

Q: A grayscale image appears dull with very little difference between object and background intensities. (a) Interpret the scenario and explain the cause of low contrast. (b) Identify all key issues affecting image visibility. (c) Apply an appropriate enhancement technique to address the problem. (d) Justify your choice with logical reasoning. (e) Present your explanation in a clear and structured manner.

A: (a) Low contrast occurs when pixel intensities are clustered in a narrow range of the grayscale spectrum, usually due to poor lighting during capture, sensor limitations, or improper exposure settings. The narrow intensity range makes objects blend visually with the background. (b) Key issues: narrow dynamic range, poor separation between object and background, loss of perceptual detail, difficulty in further processing (edge detection, segmentation). (c) Apply Histogram Equalization (or contrast stretching) to redistribute intensity values across the full 0–255 range. (d) Histogram equalization is justified because it automatically spreads out the most frequent intensity values, improving global contrast without needing manual parameter tuning, unlike simple linear stretching which may not handle skewed histograms well. (e) Structured summary: Cause → narrow intensity range → Fix → histogram equalization → Result → improved global contrast and visibility.

2. Uneven Illumination

Q: An image captured under non-uniform lighting shows bright and dark patches. (a) Interpret the scenario and explain the impact of uneven illumination. (b) Identify key factors causing intensity variation. (c) Apply a suitable enhancement method to normalize illumination. (d) Justify your selected approach with reasoning. (e) Provide a clear and well-structured explanation.

A: (a) Uneven illumination causes spatially varying brightness, making uniform thresholding or segmentation unreliable since the same object may have different intensities in different regions. (b) Factors: non-uniform light source position, shadows, reflective surfaces, sensor vignetting. (c) Apply Homomorphic Filtering, which separates illumination (low-frequency) and reflectance (high-frequency) components in the frequency domain and normalizes the illumination component. (d) Homomorphic filtering is preferred over simple contrast adjustment because it directly targets the illumination component mathematically (via log transform + frequency filtering), correcting brightness variation while preserving reflectance detail. (e) Cause → non-uniform lighting → Fix → homomorphic filtering → Result → normalized, uniformly lit image.

3. Salt-and-Pepper Noise

Q: A scanned document contains salt-and-pepper noise affecting readability. (a) Explain the scenario and characteristics of this noise. (b) Identify key issues impacting image quality. (c) Apply an appropriate filtering technique. (d) Justify why this method is more suitable than alternatives. (e) Present your response clearly and logically.

A: (a) Salt-and-pepper noise appears as randomly scattered black and white pixels, typically from faulty scanner sensors, transmission errors, or dust — it corrupts isolated pixels rather than whole regions. (b) Issues: broken text strokes, false dark/bright dots, reduced OCR/readability accuracy. (c) Apply a Median Filter. (d) The median filter replaces each pixel with the median of its neighborhood, effectively removing extreme outlier values (the noise) while preserving edges — unlike a mean filter, which would blur the noise into surrounding pixels and smear edges. (e) Cause → impulse noise on isolated pixels → Fix → median filter → Result → clean, readable document image.

4. Blurred Edges

Q: An image processed for noise removal appears overly smooth, causing edges to blur. (a) Interpret the scenario and explain why edges are blurred. (b) Identify key issues related to smoothing and edge loss. (c) Apply an appropriate technique to restore edges. (d) Justify your decision with clear reasoning. (e) Structure your explanation clearly.

A: (a) Aggressive smoothing filters (e.g., large Gaussian or mean filters) average pixel values across neighborhoods, which reduces noise but also averages out the sharp intensity transitions that define edges. (b) Issues: loss of high-frequency detail, reduced sharpness, difficulty in subsequent edge/feature detection. (c) Apply Unsharp Masking / High-Boost Filtering to re-emphasize edges. (d) Unsharp masking works by subtracting a blurred version of the image from the original to obtain edge information, then adding this back with a weight — restoring sharpness without reintroducing noise, unlike simply reducing the smoothing filter size which

reduces noise removal effectiveness. (e) Cause → over-smoothing → Fix → unsharp masking → Result → sharper edges with noise still suppressed.

5. High-Frequency Noise

Q: A satellite image contains fine-grained noise affecting feature visibility. (a) Interpret the scenario and describe high-frequency noise. (b) Identify the key issues affecting image clarity. (c) Apply an appropriate frequency domain filtering method. (d) Justify the choice of filter with reasoning. (e) Present your answer in a clear and organized manner.

A: (a) High-frequency noise manifests as fine, rapid intensity fluctuations — appearing as grainy texture — usually from sensor electronics or atmospheric interference, and it occupies the high-frequency band in the Fourier spectrum. (b) Issues: reduced feature clarity, false fine-detail artifacts, difficulty in classification/feature extraction. (c) Apply a Low-Pass Filter in the Frequency Domain (e.g., Butterworth or Gaussian low-pass filter). (d) A Butterworth low-pass filter is justified because it offers a smooth, controllable cutoff (via its order parameter), suppressing high-frequency noise while avoiding the ringing artifacts caused by an ideal low-pass filter's sharp cutoff. (e) Cause → high-frequency grain → Fix → frequency-domain low-pass filtering → Result → smoother image with retained large-scale features.

6. Object Boundary Detection

Q: In a medical image, boundaries between tissues need to be clearly identified. (a) Interpret the scenario and explain the importance of boundary detection. (b) Identify key challenges in detecting boundaries. (c) Apply a suitable edge detection method. (d) Justify your selection with logical reasoning. (e) Present your explanation clearly and systematically.

A: (a) Accurate boundary detection is critical in medical imaging for identifying organ/tissue borders, tumors, or lesions, directly impacting diagnosis and treatment planning. (b) Challenges: low contrast between tissues, noise, gradual intensity transitions rather than sharp edges. (c) Apply the Canny Edge Detector. (d) Canny is justified because it combines Gaussian smoothing (noise reduction), gradient-based edge strength/direction calculation, non-maximum suppression (thin edges), and hysteresis thresholding (connects weak-but-relevant edges) — giving more accurate, continuous, and noise-robust boundaries than simpler operators like Sobel or Prewitt. (e) Need → precise tissue boundaries → Fix → Canny edge detection → Result → clear, continuous tissue boundaries for diagnosis.

7. Simple Segmentation

Q: An image contains objects clearly distinguishable from the background based on intensity. (a) Interpret the scenario and explain segmentation requirements. (b) Identify key features enabling separation. (c) Apply a suitable segmentation technique. (d) Justify your decision logically. (e) Present your response clearly.

A: (a) Since objects and background have distinctly different intensity levels, a simple pixel-intensity-based separation is sufficient — no need for complex region- or texture-based methods. (b) Feature: a clear bimodal histogram separating object and background intensity peaks. (c) Apply Global Thresholding (e.g., Otsu's method). (d) Otsu's thresholding automatically computes the optimal threshold by maximizing between-class variance, which is efficient and accurate when the histogram is clearly bimodal, avoiding the need for manual threshold selection. (e) Feature → distinct intensity separation → Fix → global (Otsu) thresholding → Result → clean binary segmentation.

8. Complex Segmentation

Q: An image contains multiple regions with overlapping intensity values. (a) Interpret the scenario and explain segmentation challenges. (b) Identify key issues such as intensity overlap. (c) Apply an appropriate advanced segmentation method. (d) Justify how the method handles complexity. (e) Provide a structured and clear explanation.

A: (a) When intensity ranges of different regions overlap, a single global threshold cannot separate them correctly, since some background pixels may share intensity values with objects. (b) Issues: histogram has overlapping/multiple modes, ambiguous boundaries, risk of misclassification with simple thresholding. (c) Apply K-Means Clustering or Watershed Segmentation. (d) K-Means groups pixels based on similarity in a multi-dimensional feature space (intensity, position, texture), allowing it to separate overlapping regions better than a single threshold; watershed additionally uses topological/gradient information to separate touching regions — both handle complexity beyond simple intensity cut-offs. (e) Challenge → overlapping intensities → Fix → clustering/watershed → Result → better-separated multiple regions.

9. Broken Edges

Q: After applying edge detection, object boundaries appear discontinuous. (a) Interpret the scenario and explain why edges are broken. (b) Identify key issues affecting boundary continuity. (c) Apply a suitable method to improve edge connectivity. (d) Justify your approach logically. (e) Present your explanation clearly.

A: (a) Edges break when gradient magnitude fluctuates below the detection threshold at certain points, often due to noise, weak contrast at those points, or overly strict thresholding. (b) Issues: gaps in boundaries, incomplete object outlines, unreliable shape analysis. (c) Apply Morphological Dilation followed by thinning, or use Hysteresis Thresholding with edge linking (as in Canny). (d) Hysteresis thresholding is justified because it retains weak edges that are connected to strong edges, bridging small gaps naturally based on connectivity rather than blindly thickening all edges (which morphological dilation alone would do, potentially distorting shape). (e) Cause → threshold-induced gaps → Fix → hysteresis-based edge linking → Result → continuous, connected boundaries.

10. Combined Requirement

Q: An industrial inspection system requires noise removal while preserving important edges for defect detection. (a) Interpret the scenario and define the combined requirement. (b) Identify key issues involving noise and edge preservation. (c) Apply an appropriate sequence of techniques. (d) Justify the order and selection with strong reasoning. (e) Present your response in a clear and structured format.

A: (a) The system needs simultaneous noise suppression and edge preservation — a trade-off, since most smoothing filters blur edges along with noise. (b) Issues: noise obscures defects; smoothing removes noise but also erases fine defect edges needed for detection. (c) Apply a Bilateral Filter (edge-preserving smoothing) followed by Canny edge detection for defect boundary extraction. (d) The bilateral filter is applied first because it smooths flat regions using both spatial and intensity similarity, removing noise while keeping edges sharp; Canny is applied after because it works best on a cleaned image, minimizing false edges from noise while accurately capturing true defect boundaries. (e) Requirement → noise-free + edge-preserved → Fix → bilateral filter → Canny edge detection → Result → accurate defect detection pipeline.

11. Low Brightness Image

Q: An image captured in low light appears very dark with poor visibility. (a) Interpret the scenario and explain the cause of low brightness. (b) Identify key issues affecting image visibility. (c) Apply an appropriate enhancement technique. (d) Justify your choice with reasoning. (e) Present your explanation clearly.

A: (a) Low brightness results from insufficient light reaching the sensor, causing pixel intensities to cluster near the dark end of the range. (b) Issues: poor visibility of details, low contrast in shadow regions, difficulty in feature extraction. (c) Apply Gamma Correction (with $\gamma < 1$) or logarithmic transformation. (d) Gamma correction with $\gamma < 1$ expands dark intensity values non-linearly, brightening the image while compressing highlights, giving a more visually natural correction than a simple linear brightness addition, which can cause clipping. (e) Cause → insufficient light → Fix → gamma correction → Result → brighter, more visible image.

12. Overexposed Image

Q: An image appears excessively bright, losing important details. (a) Interpret the scenario and explain overexposure. (b) Identify key issues affecting intensity representation. (c) Apply a suitable correction technique. (d) Justify your choice logically. (e) Present your explanation in a structured manner.

A: (a) Overexposure occurs when too much light reaches the sensor, pushing many pixel values to saturation (near 255), causing loss of detail in bright regions. (b) Issues: clipped highlights, washed-out details, poor dynamic range utilization. (c) Apply Gamma Correction (with $\gamma > 1$) or inverse log transformation. (d) Gamma correction with $\gamma > 1$ compresses bright values and expands darker ones,

recovering some detail in highlight regions without needing to fully re-capture the image, making it more effective than simple intensity subtraction which risks losing the little remaining detail. (e) Cause → sensor saturation → Fix → gamma correction ($\gamma > 1$) → Result → recovered highlight detail.

13. Gaussian Noise

Q: An image contains Gaussian noise affecting smooth regions. (a) Interpret the scenario and describe Gaussian noise. (b) Identify key issues affecting quality. (c) Apply an appropriate smoothing filter. (d) Justify your method over alternatives. (e) Present your explanation clearly.

A: (a) Gaussian noise adds random intensity variations following a normal distribution to every pixel, typically arising from sensor thermal noise or electronic interference. (b) Issues: grainy appearance in smooth areas, reduced signal-to-noise ratio, degraded visual quality. (c) Apply a Gaussian Filter (or Wiener filter for optimal statistical noise reduction). (d) A Gaussian filter is well-matched to Gaussian noise because its weighted averaging (higher weight to central pixels) suppresses random Gaussian-distributed variations effectively while introducing less blur than a simple mean filter with uniform weights. (e) Cause → sensor-induced Gaussian noise → Fix → Gaussian smoothing filter → Result → reduced noise with preserved smooth-region quality.

14. Speckle Noise

Q: An image obtained from ultrasound imaging shows speckle noise. (a) Interpret the scenario and explain speckle noise characteristics. (b) Identify key issues affecting clarity. (c) Apply a suitable filtering technique. (d) Justify your approach logically. (e) Present your explanation clearly.

A: (a) Speckle noise is a multiplicative, granular noise pattern caused by coherent wave interference (common in ultrasound and radar imaging), unlike additive noise types. (b) Issues: grainy texture obscuring tissue boundaries, reduced diagnostic clarity, multiplicative nature makes standard additive-noise filters less effective. (c) Apply a Lee Filter or Median Filter adapted for multiplicative noise. (d) The Lee filter is justified because it adapts its smoothing based on local statistics (mean and variance), effectively handling the multiplicative nature of speckle noise better than a standard linear filter designed for additive noise. (e) Cause → coherent-wave speckle noise → Fix → Lee/median filtering → Result → clearer ultrasound image with visible tissue structure.

15. Loss of Fine Details

Q: An image loses fine details after processing. (a) Interpret the scenario and explain detail loss. (b) Identify key issues related to filtering. (c) Apply a suitable enhancement technique. (d) Justify your method with reasoning. (e) Present your explanation clearly.

A: (a) Fine detail loss typically results from aggressive low-pass filtering or compression, which removes high-frequency components that represent small-scale structures and texture. (b) Issues: blurred textures, indistinct small objects, reduced usability for detailed analysis. (c) Apply High-Pass Filtering or Unsharp Masking to re-emphasize fine detail. (d) High-pass filtering isolates and amplifies the high-frequency components that were suppressed, restoring detail without needing to redo the original processing, making it more efficient than reprocessing from scratch. (e) Cause → over-filtering removed high frequencies → Fix → high-pass/unsharp enhancement → Result → restored fine detail.

16. Edge Enhancement Requirement

Q: An application requires highlighting edges for feature extraction. (a) Interpret the scenario and explain the need for edge enhancement. (b) Identify key issues in detecting edges. (c) Apply an appropriate method. (d) Justify your choice logically. (e) Present your explanation clearly.

A: (a) Feature extraction pipelines (e.g., object recognition) rely on well-defined edges to identify shapes, so enhancing edge visibility improves downstream accuracy. (b) Issues: weak or noisy edges, inconsistent gradient strength across the image. (c) Apply the Sobel Operator (or Laplacian filter for second-derivative edge enhancement). (d) The Sobel operator is justified for feature extraction tasks because it computes directional gradients (horizontal and vertical), giving both edge strength and orientation, which is valuable input for feature descriptors, while being computationally lightweight compared to more complex detectors. (e) Need → distinct edges for features → Fix → Sobel operator → Result → enhanced, direction-aware edges.

17. Background Noise Removal

Q: An image contains background noise affecting segmentation accuracy. (a) Interpret the scenario and explain noise impact. (b) Identify key issues affecting segmentation. (c) Apply a suitable preprocessing method. (d) Justify your approach logically. (e) Present your explanation clearly.

A: (a) Background noise introduces random intensity fluctuations that can be mistaken for object boundaries or separate regions, reducing segmentation reliability. (b) Issues: false region boundaries, over-segmentation, unreliable thresholds. (c) Apply a Gaussian or Median Filter as preprocessing before segmentation. (d) Pre-smoothing with a Gaussian filter reduces random noise-induced intensity spikes while retaining overall structure, making the histogram more clearly bimodal/multimodal and improving the reliability of the subsequent segmentation step, compared to segmenting the noisy image directly. (e) Issue → noise corrupts segmentation → Fix → pre-filtering → Result → cleaner input for accurate segmentation.

18. Threshold Selection Problem

Q: An image requires segmentation, but selecting a threshold is difficult. (a) Interpret the scenario and explain the challenge. (b) Identify key issues in threshold selection. (c) Apply a suitable thresholding method. (d) Justify your approach logically. (e) Present your explanation clearly.

A: (a) Manual threshold selection is unreliable when the histogram is not clearly bimodal or when illumination varies across the image, making a single fixed threshold inadequate. (b) Issues: ambiguous histogram peaks, varying illumination, risk of under/over-segmentation with a poorly chosen threshold. (c) Apply Otsu's Method for automatic global thresholding, or Adaptive (Local) Thresholding if illumination varies spatially. (d) Otsu's method is justified because it statistically determines the threshold that minimizes intra-class variance without manual tuning; adaptive thresholding is preferred instead when illumination is non-uniform, as it computes local thresholds per region. (e) Challenge → no clear manual threshold → Fix → Otsu/adaptive thresholding → Result → objective, reliable segmentation.

19. Texture-Based Segmentation

Q: An image contains regions distinguishable by texture rather than intensity. (a) Interpret the scenario and explain segmentation challenges. (b) Identify key issues with intensity-based methods. (c) Apply an appropriate segmentation approach. (d) Justify your method logically. (e) Present your explanation clearly.

A: (a) When regions share similar average intensity but differ in texture pattern, intensity-based segmentation cannot distinguish them, requiring a texture-aware approach. (b) Issues: intensity thresholding ignores spatial pattern information, leading to failed separation of texturally distinct but intensity-similar regions. (c) Apply Texture-Based Segmentation using Gray-Level Co-occurrence Matrix (GLCM) features combined with clustering, or Gabor filter banks. (d) GLCM captures spatial relationships between pixel intensities (contrast, homogeneity, energy), providing texture descriptors that intensity histograms cannot, allowing segmentation based on pattern rather than brightness alone. (e) Challenge → texture, not intensity, differs → Fix → GLCM/Gabor-based texture segmentation → Result → regions correctly separated by texture.

20. Multiple Object Detection

Q: An image contains multiple objects with similar intensity values. (a) Interpret the scenario and explain segmentation difficulty. (b) Identify key issues in distinguishing objects. (c) Apply a suitable segmentation technique. (d) Justify your choice logically. (e) Present your explanation clearly.

A: (a) When multiple objects share similar intensity, simple thresholding merges them into one region, making individual object detection impossible without additional spatial cues. (b) Issues: touching/overlapping objects, indistinguishable intensity, risk of merging separate objects into one blob. (c) Apply Watershed Segmentation (often

combined with distance transform and marker-based control). (d) Watershed segmentation is justified because it treats the image as a topographic surface and separates touching objects along "ridge lines," using shape/distance information rather than intensity alone, effectively splitting objects that thresholding would merge. (e) Challenge → similar-intensity multiple objects → Fix → marker-based watershed → Result → individually separated objects.

21. Edge Noise Problem

Q: Edges detected are affected by noise. (a) Interpret the scenario and explain the issue. (b) Identify key challenges in edge detection. (c) Apply a suitable solution. (d) Justify your method logically. (e) Present your explanation clearly.

A: (a) Noise creates spurious high-gradient points that edge detectors misidentify as real edges, resulting in false positives and cluttered edge maps. (b) Challenges: false edges from noise, difficulty distinguishing true object boundaries from noise-induced gradients. (c) Apply Gaussian smoothing before edge detection (as in the Canny pipeline), or use the Canny detector directly, which integrates this step. (d) Pre-smoothing with a Gaussian filter reduces small random fluctuations before gradient computation, so only genuine, significant intensity transitions are detected as edges, which is more reliable than detecting edges on the raw noisy image. (e) Issue → noise-induced false edges → Fix → Gaussian smoothing + Canny → Result → cleaner, accurate edge map.

22. Region Merging Requirement

Q: An image contains fragmented regions that need merging. (a) Interpret the scenario and explain the issue. (b) Identify key challenges in region segmentation. (c) Apply an appropriate technique. (d) Justify your approach logically. (e) Present your explanation clearly.

A: (a) Fragmentation happens when a single object is split into multiple small regions due to noise, texture variation, or over-sensitive segmentation parameters. (b) Challenges: excessive small regions, loss of object-level coherence, harder subsequent analysis. (c) Apply Region Merging (post-processing step in region-based segmentation, merging adjacent regions with similar mean intensity/color below a similarity threshold). (d) Region merging is justified because it directly targets the over-fragmentation by combining statistically similar adjacent regions, restoring meaningful object-level segments rather than requiring re-segmentation with different parameters. (e) Issue → fragmented regions → Fix → similarity-based region merging → Result → coherent, merged object regions.

23. Over-Segmentation

Q: An image is divided into too many regions after segmentation. (a) Interpret the scenario and explain over-segmentation. (b) Identify key issues affecting segmentation. (c) Apply a suitable correction method. (d) Justify your choice logically. (e) Present your explanation clearly.

A: (a) Over-segmentation occurs when algorithms (commonly watershed) are overly sensitive to local minima/noise, producing many small, meaningless regions instead of clean object boundaries. (b) Issues: excessive fragmentation, loss of meaningful structure, higher computational cost for downstream analysis. (c) Apply Marker-Controlled Watershed Segmentation or region merging as post-processing. (d) Marker-controlled watershed restricts flooding to start only from selected significant markers (e.g., local maxima of a distance transform), reducing the number of unnecessary basins/regions compared to standard watershed, directly addressing the root cause. (e) Issue → excessive regions → Fix → marker-controlled watershed → Result → meaningful, reduced region count.

24. Under-Segmentation

Q: Important regions are merged incorrectly during segmentation. (a) Interpret the scenario and explain under-segmentation. (b) Identify key issues affecting separation. (c) Apply a suitable solution. (d) Justify your approach logically. (e) Present your explanation clearly.

A: (a) Under-segmentation happens when distinct regions are incorrectly merged, usually due to weak boundaries, insufficient contrast, or overly coarse segmentation parameters. (b) Issues: loss of important object separation, incorrect region statistics, poor object-level analysis. (c) Apply Finer Thresholding/Adaptive Segmentation, or add boundary/edge constraints (e.g., combine region growing with edge information). (d) Incorporating edge information alongside region-based segmentation ensures that region growth stops at true boundaries, preventing merging across genuine object edges — more effective than adjusting a single global parameter which may not generalize. (e) Issue → incorrect merging of regions → Fix → edge-constrained/adaptive segmentation → Result → correctly separated important regions.

25. Boundary Refinement

Q: Detected boundaries are rough and inaccurate. (a) Interpret the scenario and explain the issue. (b) Identify key challenges in boundary detection. (c) Apply a suitable refinement method. (d) Justify your choice logically. (e) Present your explanation clearly.

A: (a) Rough boundaries result from noise, coarse segmentation, or discretization artifacts, producing jagged rather than smooth, accurate object outlines. (b) Challenges: jagged/staircase edges, small protrusions or gaps, inaccurate shape representation. (c) Apply Active Contour Models (Snakes) or Morphological Smoothing (opening/closing) for boundary refinement. (d) Active contours are justified because they iteratively deform a boundary curve to minimize an energy function balancing smoothness and fit to image gradients, producing refined, smooth, and accurate boundaries beyond what simple morphological operations achieve. (e) Issue → rough boundaries → Fix → active contour refinement → Result → smooth, accurate object boundaries.

26. Frequency-Based Enhancement

Q: An image requires enhancement using frequency domain methods. (a) Interpret the scenario and explain frequency-based processing. (b) Identify key issues in spatial domain methods. (c) Apply an appropriate frequency domain technique. (d) Justify your choice logically. (e) Present your explanation clearly.

A: (a) Frequency-domain processing transforms the image (via Fourier Transform) into frequency components, allowing targeted manipulation of specific frequency bands (e.g., removing periodic noise, enhancing edges) that may be harder to isolate spatially. (b) Issues with spatial methods: convolution-based filters can be computationally expensive for large kernels, and it's harder to isolate specific periodic/structured noise patterns directly in the spatial domain. (c) Apply Fast Fourier Transform (FFT) based filtering, such as a Band-Reject/Notch Filter for periodic noise or a High/Low-Pass Filter for general enhancement. (d) Frequency domain filtering is justified when noise or the enhancement goal corresponds to a specific frequency signature, making it easy to precisely target and remove/enhance via a notch or band filter — something difficult to achieve with a spatial kernel. (e) Need → targeted frequency-based control → Fix → FFT-based filtering → Result → enhanced image with targeted frequency correction.

27. Mixed Noise Scenario

Q: An image contains both Gaussian and impulse noise. (a) Interpret the scenario and explain challenges. (b) Identify key issues affecting filtering. (c) Apply a suitable combined approach. (d) Justify your method logically. (e) Present your explanation clearly.

A: (a) Mixed noise combines random additive Gaussian variations with sudden extreme impulse (salt-and-pepper) values, and no single filter type optimally handles both simultaneously. (b) Issues: median filters handle impulse noise well but don't optimally reduce Gaussian noise; Gaussian filters smooth Gaussian noise but don't remove impulse spikes effectively (they blur them instead of removing). (c) Apply a Sequential/Hybrid Filtering Approach: Median filter first (to remove impulse noise), followed by a Gaussian or Wiener filter (to reduce residual Gaussian noise). (d) Applying the median filter first removes the extreme outlier values without them corrupting the statistics used by the subsequent Gaussian/Wiener filter, resulting in more effective overall noise reduction than either filter used alone. (e) Challenge → two noise types present → Fix → median filter then Gaussian/Wiener filter → Result → noise from both sources reduced.

28. Edge Preservation Filtering

Q: A filtering method is required to remove noise but preserve edges. (a) Interpret the scenario and explain the requirement. (b) Identify key issues in filtering. (c) Apply a suitable technique. (d) Justify your approach logically. (e) Present your explanation clearly.

A: (a) The requirement is to denoise flat/smooth regions without blurring true object boundaries — a common trade-off in image restoration. (b) Issues: conventional linear filters (mean/Gaussian) smooth indiscriminately, blurring edges along with noise. (c) Apply a Bilateral Filter or Anisotropic Diffusion (Perona-Malik). (d) The bilateral filter weights neighboring pixels by both spatial distance and intensity similarity, avoiding averaging across edges, while anisotropic diffusion halts diffusion across high-gradient boundaries — both address edge preservation better than standard smoothing filters. (e) Requirement → denoise without blurring edges → Fix → bilateral filter/anisotropic diffusion → Result → clean image with sharp edges retained.

29. Region Growing Failure

Q: Region growing fails due to incorrect seed selection. (a) Interpret the scenario and explain the issue. (b) Identify key challenges in region growing. (c) Apply a suitable correction. (d) Justify your method logically. (e) Present your explanation clearly.

A: (a) Region growing segments an image by expanding from initial seed points based on similarity criteria; if seeds are placed on noise, boundaries, or non-representative pixels, the algorithm grows incorrect or incomplete regions. (b) Challenges: seed sensitivity, propagation of errors from a poor starting point, inconsistent results across runs. (c) Apply Automatic Seed Selection using local intensity maxima/minima or clustering (e.g., using histogram peaks) instead of manual/arbitrary seed placement. (d) Automatic seed selection based on statistically representative points ensures seeds start within homogeneous, representative regions, reducing the risk of incorrect growth compared to manual or random seed choice. (e) Issue → poor seed choice → Fix → automatic, criteria-based seed selection → Result → reliable region growing outcome.

30. Industrial Inspection Enhancement

Q: An industrial system requires detecting defects in noisy images. (a) Interpret the scenario and define requirements. (b) Identify key issues in detection. (c) Apply an appropriate enhancement and segmentation pipeline. (d) Justify your sequence logically. (e) Present your explanation clearly.

A: (a) The system must reliably detect small, potentially subtle defects despite the presence of noise, meaning the pipeline must denoise without destroying defect-relevant detail, then segment/localize defects accurately. (b) Issues: noise can mask small defects or create false positives; over-smoothing can erase real defects; poor segmentation can miss or merge defect regions with background. (c) Apply a pipeline: (1) Median/Bilateral Filtering (noise removal with edge preservation) → (2) Contrast Enhancement (histogram equalization) → (3) Edge Detection (Canny) or Thresholding/Watershed Segmentation to isolate defect regions. (d) The sequence is justified because denoising must happen first to prevent noise corrupting later contrast/edge steps; contrast enhancement next makes subtle defects distinguishable; segmentation is applied last since it depends on a clean, enhanced

image — reversing this order would let noise or poor contrast degrade the final segmentation accuracy. (e) Requirement → reliable defect detection in noisy images → Fix → filter → enhance → segment pipeline → Result → accurate, robust defect detection.

Code of Conduct:

I K.L.V.S Karthik (192424177) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation